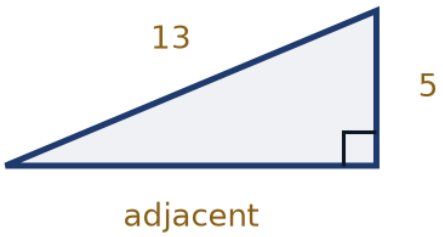
# Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

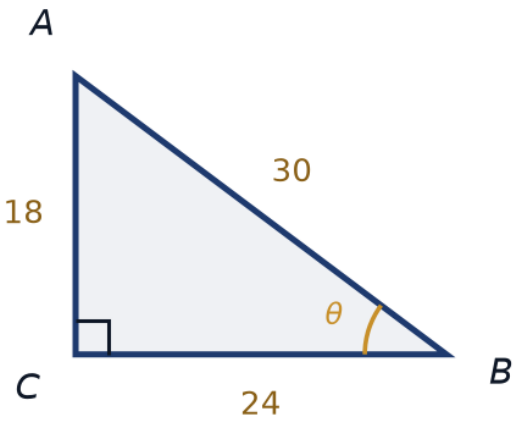
**1** [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle $\theta$ has an opposite side of $5$ and hypotenuse $13$. Find $\sin\theta$ and the length of the adjacent side.




**2** [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle $\theta$ has an opposite side of $15$ and hypotenuse $17$. Find $\cos\theta$.




**3** [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle $\theta$ has an adjacent side of $7$ and hypotenuse $25$. Find $\sin\theta$ and $\tan\theta$.

**4** [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, the two legs are $20$ and $21$, and the hypotenuse is $29$. Find $\sin\theta$ and $\cos\theta$ for the angle $\theta$ opposite the side of length $20$.

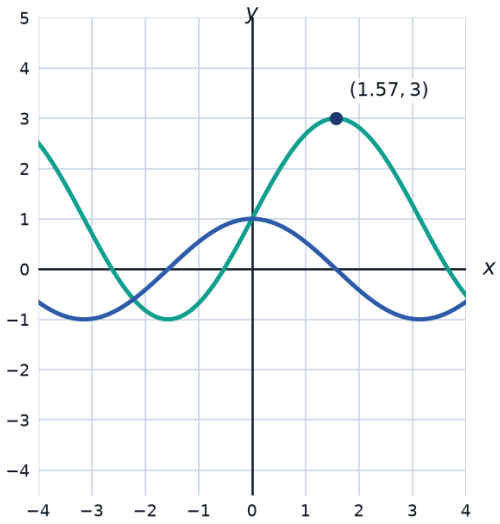
**5** [Foundational] • Target: Special angles

State the exact value of $\sin(30^\circ)$.

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**6** [Foundational] • Target: Special angles

State the exact value of $\cos(45^\circ)$.




**7** [Foundational] • Target: Special angles

State the exact value of $\tan(60^\circ)$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Special angles

State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.

**9** [Intermediate] • Target: Converting between radians and degrees

Convert $180^\circ$ to radians.

**10** [Intermediate] • Target: Converting between radians and degrees

Convert $\frac{5\pi}{6}$ radians to degrees.

**11** [Intermediate] • Target: The Pythagorean identity

Given $\sin\theta = \frac{3}{5}$ with $\theta$ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

**12** [Intermediate] • Target: The Pythagorean identity

Given $\cos\theta = \frac{5}{13}$ with $\theta$ acute, find $\sin\theta$.

**13** [Intermediate] • Target: The Pythagorean identity

Given $\sin\theta = 0.6$, find $\cos^2\theta$.

**14** [Intermediate] • Target: Simplifying trigonometric expressions

Simplify $\sin^2\theta + \cos^2\theta + 1$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Simplifying trigonometric expressions • Cross-Lesson: 2.01 Functions & Transformations

Simplify $\frac{1-\cos^2\theta}{\sin\theta}$ (for $\sin\theta \neq 0$).

**16** [Challenging] • Target: Simplifying trigonometric expressions • Cross-Lesson: 3.01 Sequences & Series

Simplify $\tan\theta\cos\theta$ (for $\cos\theta \neq 0$).

**17** [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

State the amplitude and period of $y = 5\sin(x)$.

**18** [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 6.01 Limits & Derivatives

State the amplitude and period of $y = 3\cos(2x)$.

**19** [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 7.01 Integration & Applications

State the amplitude and period of $y = 2\sin\left(\frac{x}{3}\right)$.

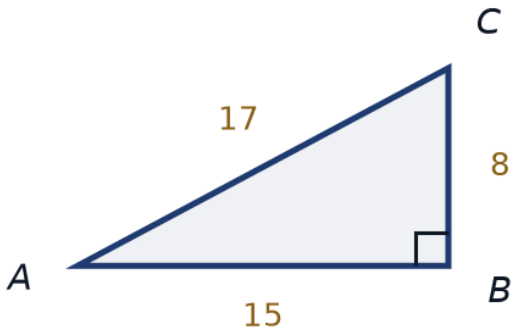
**20** [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 8.01 Exponentials & Logarithms

A sine graph has amplitude $4$ and period $\pi$. Write its equation in the form $y = A\sin(Bx)$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Right-triangle trigonometric ratios

In the right triangle shown, angle $\theta$ has opposite side $8$ and hypotenuse $17$. Find $\sin\theta$.




**a.** $\frac{17}{8}$

**b.** $\frac{8}{15}$

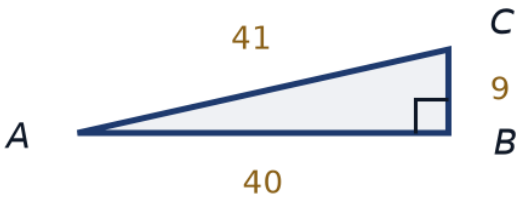
**c.** $\frac{15}{17}$

**d.** $\frac{8}{17}$

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**22** [Foundational] • Target: Right-triangle trigonometric ratios

In the right triangle shown, the two legs are $9$ and $40$, and the hypotenuse is $41$. Find $\sin\theta$ and $\cos\theta$ (where $\theta$ is opposite the side of length $9$).




**a.** $\sin\theta = \frac{9}{40}$, $\cos\theta = \frac{40}{9}$

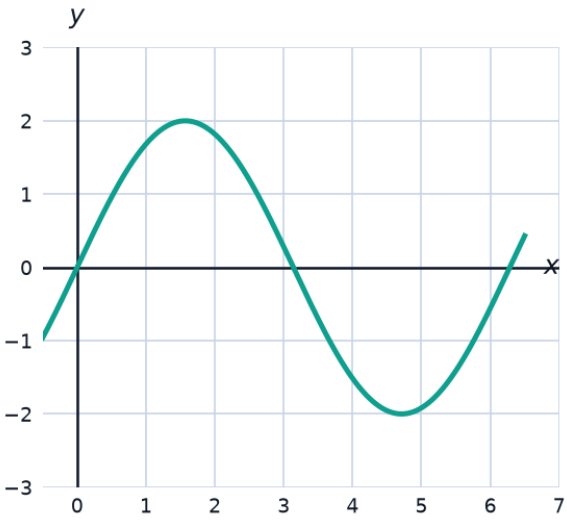
**b.** $\sin\theta = \frac{40}{41}$, $\cos\theta = \frac{9}{41}$

**c.** $\sin\theta = \frac{41}{9}$, $\cos\theta = \frac{41}{40}$

**d.** $\sin\theta = \frac{9}{41}$, $\cos\theta = \frac{40}{41}$

**23** [Foundational] • Target: Graphs: amplitude and period

The graph shows $y = A\sin(x)$ for some $A$. Find $A$ by reading the graph's amplitude.




**a.** $A = 1$

**b.** $A = 2\pi$

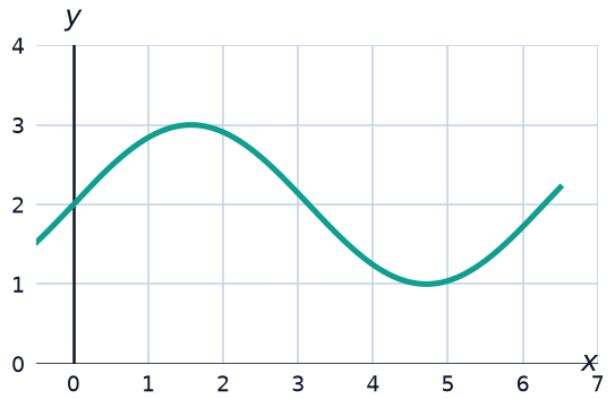
**c.** $A = 4$

**d.** $A = 2$

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**24** [Foundational] • Target: Graphs: phase shift and vertical shift

The graph shows $y = \sin(x)$ shifted vertically. Find the new midline by reading the graph.




**a.** $y = 0$

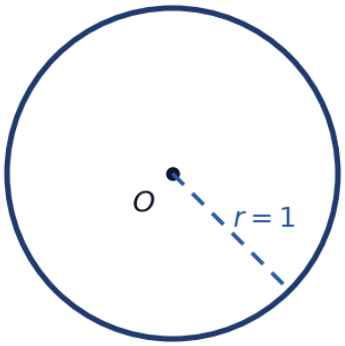
**b.** $y = 2$

**c.** $y = 1$

**d.** $y = -2$

**25** [Foundational] • Target: The unit circle: coordinates of key points

The unit circle shown marks a point at angle $\theta = 60^\circ$. What are its coordinates?




**a.** $\left(\frac{\sqrt{3}}{2}, 0.5\right)$

**b.** $(0.5, 0.5)$

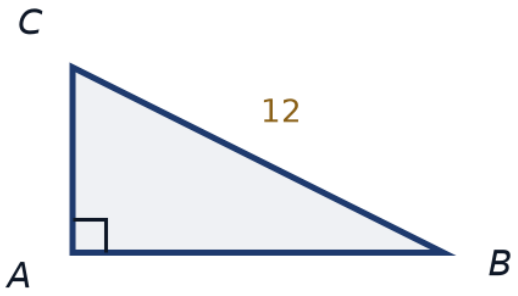
**c.** $(1, 0)$

**d.** $\left(0.5, \frac{\sqrt{3}}{2}\right)$

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**26** [Foundational] • Target: Word problems: angles of elevation

A ladder leans against a wall, making an angle of $70^\circ$ with the ground. If the ladder shown is 12 meters long, find how high up the wall it reaches, to the nearest meter.




**a.** 4

**b.** 12

**c.** 8

**d.** 11

**27** [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle $\theta$ has opposite side $9$ and hypotenuse $41$. Find $\cos\theta$.

**a.** $\frac{9}{41}$

**b.** $\frac{9}{40}$

**c.** $\frac{41}{40}$

**d.** $\frac{40}{41}$

**28** [Foundational] • Target: Right-triangle trigonometric ratios

In a right triangle, angle $\theta$ has adjacent side $20$ and hypotenuse $29$. Find $\sin\theta$.

**a.** $\frac{21}{20}$

**b.** $\frac{20}{29}$

**c.** $\frac{21}{29}$

**d.** $\frac{29}{21}$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Right-triangle trigonometric ratios

In a right triangle, angle $\theta$ has opposite side $7$ and adjacent side $24$. Find $\tan\theta$.

**a.** $\frac{24}{7}$

**b.** $\frac{24}{25}$

**c.** $\frac{7}{25}$

**d.** $\frac{7}{24}$

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**30** [Intermediate] • Target: Special angles

State the exact values of $\cos(90^\circ)$ and $\sin(0^\circ)$.

**a.** $\cos(90^\circ) = 0, \sin(0^\circ) = 1$

**b.** $\cos(90^\circ) = 1, \sin(0^\circ) = 0$

**c.** Both equal $1$

**d.** Both equal $0$

**31** [Intermediate] • Target: The Pythagorean identity

Given $\sin\theta = \frac{4}{5}$ with $\theta$ acute, find $\cos\theta$.

**a.** $\frac{3}{5}$

**b.** $\frac{4}{5}$

**c.** $\frac{5}{3}$

**d.** $\frac{9}{25}$

**32** [Intermediate] • Target: Simplifying trigonometric expressions

Simplify $\sin^2\theta + \cos^2\theta - 1$.

**a.** $2\sin^2\theta$

**b.** $0$

**c.** $2$

**d.** $1$

**33** [Intermediate] • Target: Simplifying trigonometric expressions

Simplify $\frac{1-\sin^2\theta}{\cos\theta}$ (for $\cos\theta \neq 0$).

**a.** $\frac{1}{\cos\theta}$

**b.** $\cos\theta$

**c.** $1 - \cos\theta$

**d.** $\sin\theta$

**34** [Intermediate] • Target: Simplifying trigonometric expressions

Simplify $\frac{\sin\theta}{\tan\theta}$ (for $\tan\theta \neq 0$).

**a.** $1$

**b.** $\sin\theta$

**c.** $\cos\theta$

**d.** $\frac{1}{\cos\theta}$

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**35** [Intermediate] • Target: Simplifying trigonometric expressions

Simplify $2\sin^2\theta + 2\cos^2\theta$.

**a.** 1

**b.** 0

**c.** 2

**d.** $4\sin\theta\cos\theta$

**36** [Intermediate] • Target: Graphs: amplitude and period

A cosine graph has amplitude 3 and period $4\pi$. Write its equation in the form $y = A\cos(Bx)$.

**a.** $y = 3\cos(4\pi x)$

**b.** $y = 4\pi\cos(3x)$

**c.** $y = 3\cos\left(\frac{x}{2}\right)$

**d.** $y = 3\cos(2x)$

**37** [Intermediate] • Target: Graphs: phase shift and vertical shift

The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the right and 3 units up. Write the new equation.

**a.** $y = \cos(x) + \frac{\pi}{4} + 3$

**b.** $y = \cos\left(x - \frac{\pi}{4}\right) + 3$

**c.** $y = \cos\left(x + \frac{\pi}{4}\right) + 3$

**d.** $y = \cos\left(x - \frac{\pi}{4}\right) - 3$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Graphs: phase shift and vertical shift • Cross-Lesson: 6.01 Limits & Derivatives

State the vertical shift and midline of $y = 2\sin(x) - 4$.

**a.** Down 4, midline $y = 2$

**b.** Up 4, midline $y = 4$

**c.** Down 4, midline $y = -4$

**d.** Down 2, midline $y = -4$

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**39** [Challenging] • Target: Graphs: phase shift and vertical shift • Cross-Lesson: 7.01 Integration & Applications

The graph of $y = \sin(x)$ is shifted $\frac{\pi}{6}$ to the left. Write the new equation.

**a.** $y = \sin\left(\frac{\pi}{6} - x\right)$

**b.** $y = \sin(x) + \frac{\pi}{6}$

**c.** $y = \sin\left(x + \frac{\pi}{6}\right)$

**d.** $y = \sin\left(x - \frac{\pi}{6}\right)$

**40** [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 8.01 Exponentials & Logarithms

Solve $2\sin\theta = \sqrt{2}$ for $0^\circ \le \theta < 360^\circ$.

**a.** $\theta = 45^\circ$ only

**b.** $\theta = 45^\circ$ or $\theta = 135^\circ$

**c.** $\theta = 45^\circ$ or $\theta = 225^\circ$

**d.** $\theta = 60^\circ$ or $\theta = 120^\circ$

**41** [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 9.01 Statistics & Probability

Solve $\sqrt{3}\tan\theta = 3$ for $0^\circ \le \theta < 360^\circ$.

**a.** $\theta = 60^\circ$ or $\theta = 300^\circ$

**b.** $\theta = 30^\circ$ or $\theta = 210^\circ$

**c.** $\theta = 60^\circ$ or $\theta = 240^\circ$

**d.** $\theta = 60^\circ$ only

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**42** [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 1.01 Algebra & Equations

Solve $2\cos\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

**a.** $\theta = 30^\circ$ or $\theta = 330^\circ$

**b.** $\theta = 60^\circ$ or $\theta = 240^\circ$

**c.** $\theta = 60^\circ$ only

**d.** $\theta = 60^\circ$ or $\theta = 300^\circ$

**43** [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 2.01 Functions & Transformations

Solve $\tan\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

**a.** $\theta = 45^\circ$ or $\theta = 225^\circ$

**b.** $\theta = 45^\circ$ only

**c.** $\theta = 45^\circ$ or $\theta = 315^\circ$

**d.** $\theta = 45^\circ$ or $\theta = 135^\circ$

**44** [Challenging] • Target: Solving basic trigonometric equations • Cross-Lesson: 3.01 Sequences & Series

Solve $2\sin\theta + 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

**a.** $\theta = 210^\circ$ or $\theta = 300^\circ$

**b.** $\theta = 150^\circ$ or $\theta = 330^\circ$

**c.** $\theta = 30^\circ$ or $\theta = 150^\circ$

**d.** $\theta = 210^\circ$ or $\theta = 330^\circ$

**45** [Challenging] • Target: Complementary angle relationships • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Given $\cos(40^\circ) \approx 0.766$, find $\sin(50^\circ)$.

**a.** 0.643

**b.** 0.766

**c.** 1.306

**d.** 0.234

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